Connecting the agent-based model to the data

AMR Hub

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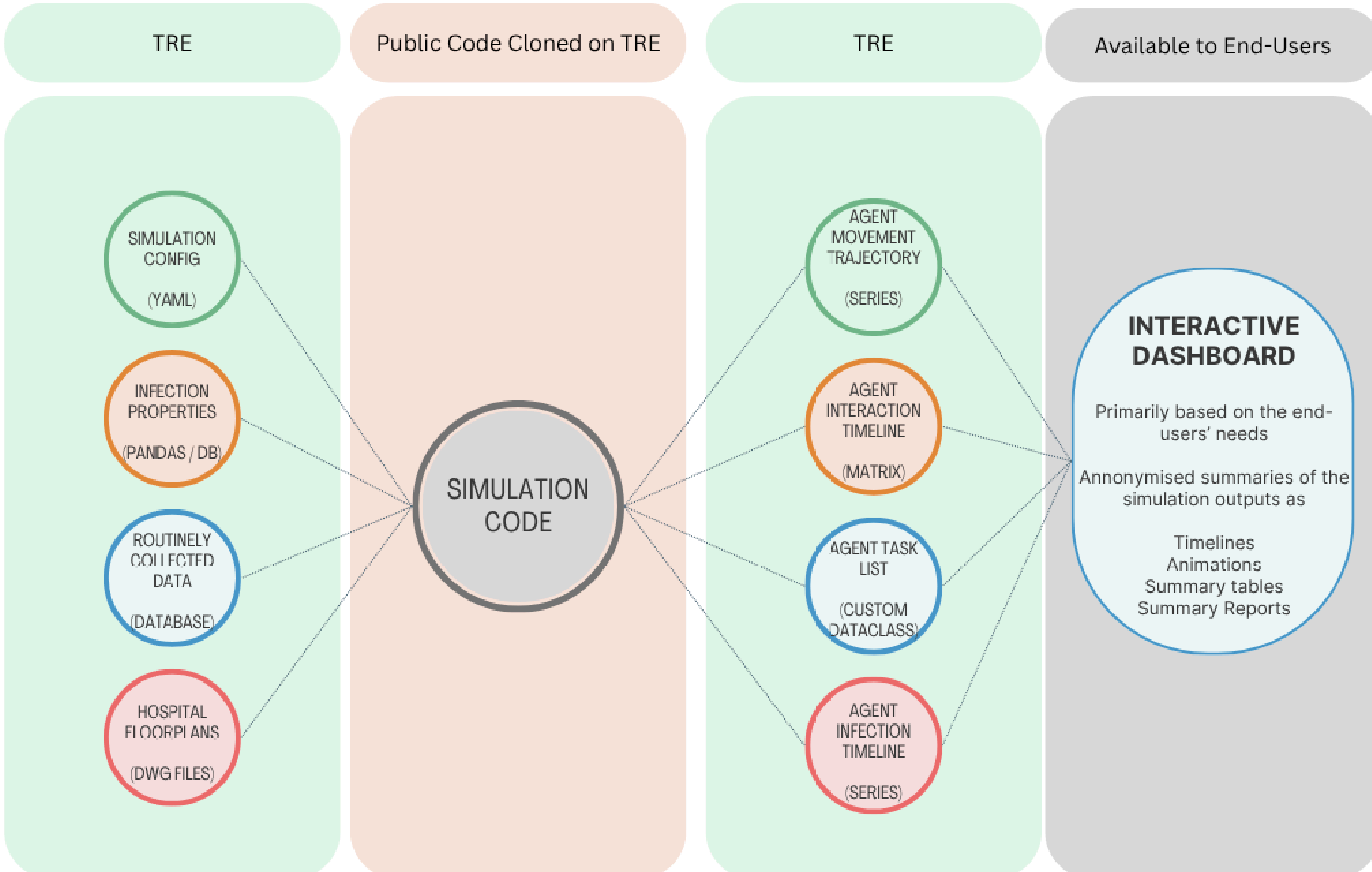
The ultimate goal



From the toy model to real data

- In the last TI, we worked heavily on the **toy model**. In principle, it can simulate agents moving through a hospital based on RCD and an environment.
- We also prepared the toy model to use **GPUs**, if and when needed.
- Towards the end of the last TI, our entire team gained access to the **actual data in the TRE**.
- We have now characterised the data to some extent and are moving towards using **IRL to parameterise the model**.

Routinely collected data: event metadata

`Data.StaffLocationEvent` — one row per recorded staff activity

Column	What it records	How we interpret it
`MasterIndexId`	Staff identifier	Joins to `Ref.Staff` for role, staff group and seniority attributes
`EventDateTime`	Event timestamp	Orders activities within each staff member’s shift
`InteractionTypeKey`	Type of recorded event	Joins to `Ref.InteractionType` to identify the activity and its class
`SourceKey`	Source object involved	A polymorphic key: resolves to a door, roster unit, workstation or bed/department according to the interaction class
`LinkKey`	Link between related events	Pairs roster-start and roster-end events to reconstruct a shift
`PatientDurableKey`	Patient identifier, when applicable	Connects patient-related activity; null for events without a patient context

Interpreting `SourceKey` by task type

Task / event type	`SourceKey` interpretation	Reliability and caveat	Most useful for
Roster	Key in `Ref.Roster` → rostered unit	**High for shift context:** paired start/end events are consistent. The unit is not a precise physical location.	Shift boundaries, unit and working context
Flowsheet	Department or bed key → room via `Ref.Department`	**Strongest room anchor:** links activity to a bed/room. Documentation may be delayed and is still a proxy for presence.	Patient-care context and room-level location
Door Message	Key in `Ref.Door` → doorway between two spaces	**Medium:** identifies a badge interaction at an edge, but not direction or confirmed passage without surrounding events.	Constraining possible movement between adjacent spaces
Workstation	Key in `Ref.Workstation` → named device	**Variable:** identifies the device reliably, but not necessarily its room; some devices are mobile. Location must be validated or inferred.	Digital task/activity evidence; location when mapping confidence is sufficient

Connecting observations to the model

Inputs define the model

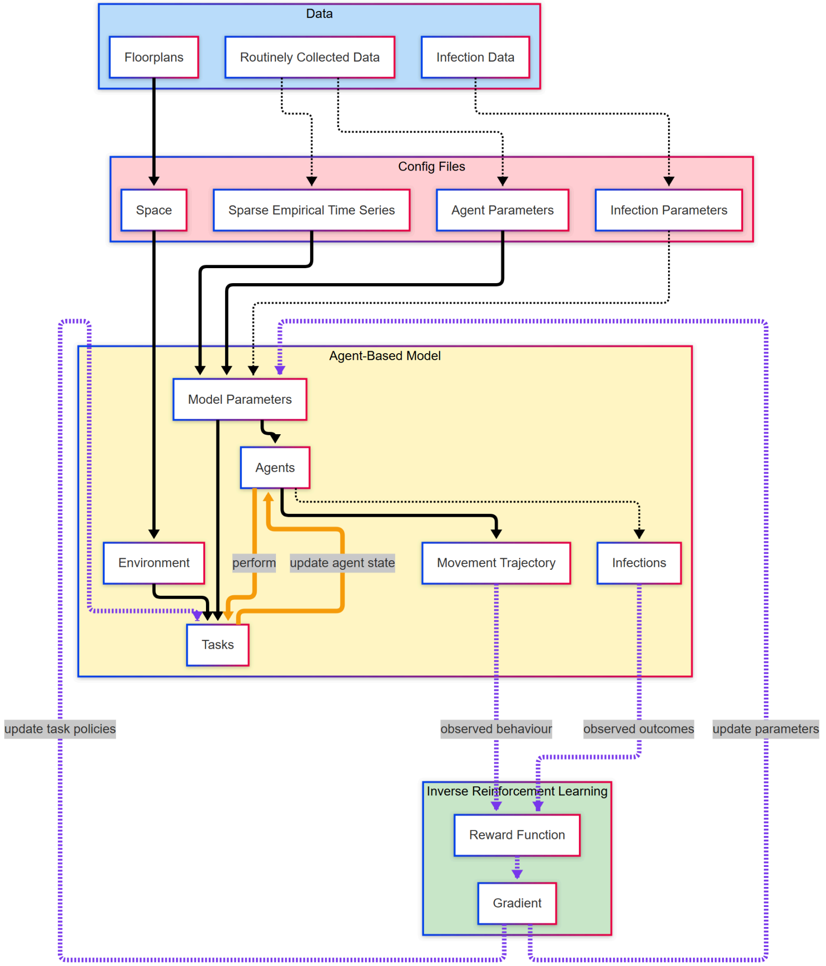
- Floorplans → spatial environment
- RCD → checkpoints and agent parameters
- Infection data → infection parameters

The ABM generates evidence

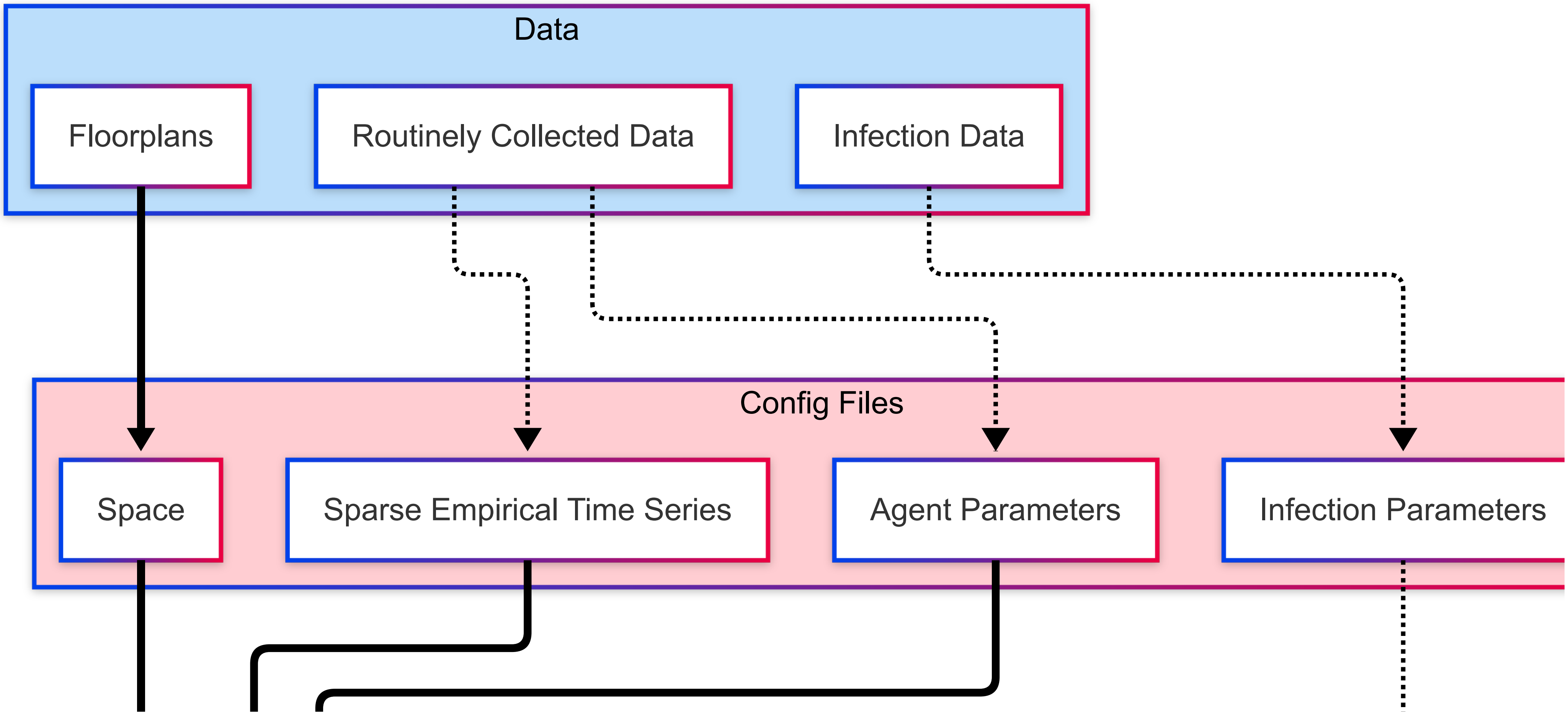
- Agents perform tasks
- Behaviour produces movement trajectories
- Interactions produce infection outcomes

IRL closes the loop

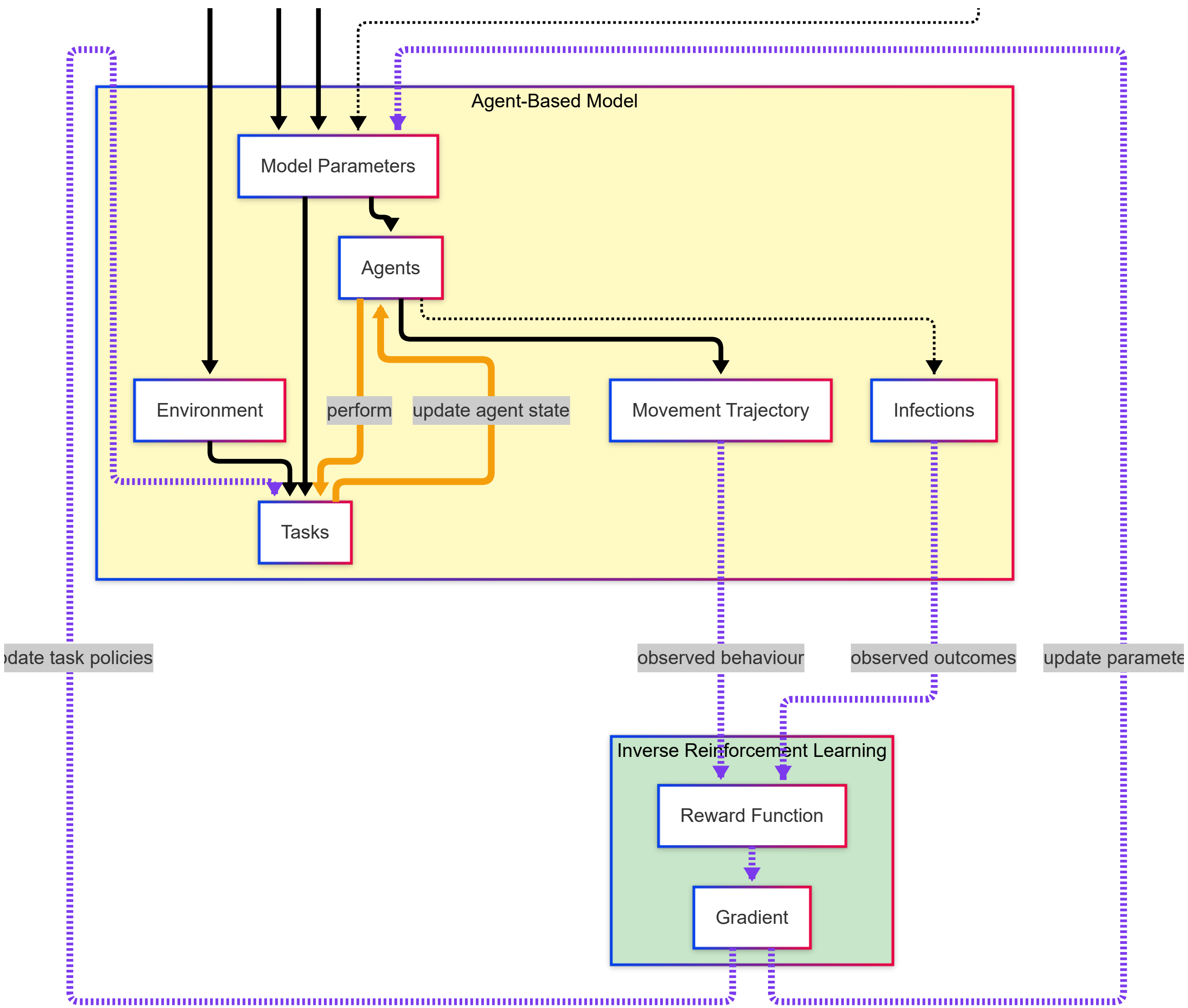
- Observations inform a reward model
- Learned gradients update task policies and parameters
- Dashed paths are proposed, not yet implemented



Zoom: inputs to the model



Zoom: parameterising the model



Proposed next steps

1. **Adapt the configuration structure**
   Integrate RCD-derived inputs cleanly into the current configuration files.
2. **Add infection to the simulation**
   Implement infection states, transmission parameters and observable outcomes.
3. **Build a canonical spatial map**
   Reconcile floorplan rooms and doors with departments, roster units and workstations.
4. **Audit the real event stream**
   Quantify coverage, shift integrity, timing delays and variation within the TRE.
5. **Construct uncertain trajectories**
   Combine spatial evidence while retaining ambiguous observations explicitly.
6. **Test whether behaviour is learnable**
   Compare contextual models with majority and Markov baselines on held-out shifts.
7. **Define the IRL problem**
   Agree states, actions, transitions, reward features and constraints.
8. **Compute the reward and implement IRL**
   Fit the reward model and connect learned behaviour to task policies and parameters.
9. **Benchmark, integrate and validate**
   Compare with simpler policies, integrate justified parameters, and test sensitivity and external validity.